

Consistency Model of the System

What CAP theorem says

In a distributed system you can only have **two of three**:

- **C** – Consistency
- **A** – Availability
- **P** – Partition tolerance

In real systems:

P is mandatory (networks can fail).

System is **CP**.

means **Consistency over Availability**.

What about writes during network or storage problem?

Event not persisted, that is: **The system becomes unavailable rather than inconsistent.**

What about reads?

Eventually consistent.

Consistency Model:

The Event-Driven Ledger system uses an **event-sourced consistency** model in which the Event Store represents the single source of financial truth. All wallet state changes, including lifecycle transitions and monetary operations, are recorded as immutable events in the Event Store and are **considered strongly consistent once persisted**.

Read-side data stores, such as the Balance and History databases, are built **asynchronously** by consuming events from the message broker. These databases are therefore **eventually consistent** with the Event Store and **may temporarily lag behind the latest written events**.